BTS–V(S)–06–24–2683 Reg. No. | | | | | | | | |

B

B.Tech. Degree V Semester Supplementary Examination June 2024

CE/CS/EC/IT/ME/EE/SE 19-200-0501 NUMERICAL AND STATISTICAL METHODS

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Solve algebraic and transcendental equations by numerical methods.
CO2: Solve numerical differentiation and integration problems.
CO3: Compute the mean and variance of a probability distribution including the binomial distribution.
CO4: Test hypotheses on data.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

			($8 \times 3 = 24$)	Marks	BL	CO	PO
I.	(a)		Prove that with usual notations (i) $hD = -\log(1-\nabla)$ (ii) $\Delta = \frac{1}{2}\delta^2 + \delta\sqrt{1+\frac{\delta^2}{4}}$.	3	L1, L3	1	1
	(b)		Given the values X: 5, 7, 11, 13, 17 f(x): 150, 392, 1452, 2366, 5202 Evaluate f(9)using. Newtons divided difference formula.	3	L3	1	1
	(c)		Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using simpsons $\frac{1}{3}$rd rule taking $h=\frac{1}{4}$.	3	L1, L3	1	3
	(d)		Use Euler modify method to find y for x = 0.1 given $\frac{dy}{dx}=\frac{y-x}{y+x}, y(0)=1$.	3	L3	2	2
	(e)		A random variable X has the probability distribution x: 0, 1, 2, 3 p(x): $\frac{k}{2}$, $\frac{k}{3}$, $\frac{k+1}{3}$, $\frac{2k-1}{6}$ Find k, Also find mean and varience of the distribution..	3	L3	3	1
	(f)		A product is 0.5% defective and is packed in cartons of 100. What % contains not more than 3 defectives?	3	L3, L4	3	3
	(g)		Define: (i) parameter and statistic (ii) critical region (iii) test statistic.	3	L1	4	1
	(h)		A random sample of 900 members has a mean 3.4 cm. Can it be reasonably regarded as a sample from a large population of mean 3.2 cm and SD 2.3 cm.	3	L4	4	4

PART B

(4 × 12 = 48) | Marks | BL | CO | PO

II. (a) Find the root of the equation $xe^x = \cos x$ using Regula- Falsi method correct to 4 decimal places. | 6 | L3 | 1 | 2

(b) Solve the system of equation by Gauss Seidel method: | 6 | L3 | 1 | 2

(i) $5x+2y+z=12$

(ii) $x+4y+2z=15$

(iii) $x+2y+5z=20$.

OR

III. (a) Using Lagrange's interpolation method find the value of f(x) at x = 5 given the values. | 6 | L3 | 1 | 2

x:	1	3	4	6
fx:	3	9	30	132

(b) Given f (40) = 184, f(50) = 204 , f(60) = 226, f(70) = 250 , f(80) = 276, f(90) = 304. Find f(38) and f (85) using suitable interpolation formula. | 6 | L3 | 1 | 3

IV. (a) Using Runge – Kutta method, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2$. | 6 | L3 | 2 | 2

(b) A slider in a machine moves along a fixed straight rod. Its distance x cm along the rod is given below for various values of the time 't'seconds. Find the velocity of the slider and it's acceleration when t = 0.3 seconds. | 6 | L2, L3 | 2 | 3

t:	0	0.1	0.2	0.3	0.4	0.5	0.6
x:	30.13	31.62	32.87	33.64	33.95	33.81	33.24

OR

V. (a) Apply Euler modified method find y(0.2) and y(0.4) given $y(0) = 0,\ y' = y + e^x$. | 6 | L2, L3 | 2 | 2

(b) The velocity 'v' (km/min) of a moped which starts from rests, is given at fixed intervals of time 't' (min) as follows: | 6 | L3 | 2 | 4

t:	2	4	6	8	10	12	14	16	18	20
x:	10	18	25	29	32	20	11	5	2	0

Evaluate approximately the distance covered in 20 minutes.

VI. (a) Derive mean and variance of Poisson Distribution. | 6 | L1 | 3 | 1

(b) Phone calls arrive at the rate of 48 per hour at the reservation desk for Indian Airlines. | 6 | L4 | 3 | 3

(i) Compute the probability of receiving three calls in a 5 minutes interval of time.

(ii) Compute the probability of receiving exactly 10 calls in 15 minutes.

(iii) Suppose no calls are currently on hold if the agent takes 5 minutes to complete the current call, how many do you expect to be waiting by that time? What is the probability that none will be waiting?

OR

VII. (a) Fit a parabola $y = a + bx + cx^2$ to the following data: | 6 | L3 | 3 | 2

x:	1	3	5	7	9
y:	3	13	31	57	91

(b) In a Normal distribution 17% of the items are below 30 and 17% of the item above 60. Find the mean and standard deviation. | 6 | L3, L4 | 3 | 3

(Continued)

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			Marks	BL	CO	PO
VIII.	(a)	It is claimed that sample of 100 tyres with mean life of 15269 km. is drawn from a population of tyres which has a mean life of 15200 km and standard deviation of 1248 km. Test the validity of the claim.	6	L4, L5	4	4
	(b)	The average marks in Mathematics of a sample of 100 students was 51 with a standard deviation of 6 marks. Could this have been a random sample from a population with average marks 50?	6	L4	4	4
		OR				
IX.	(a)	A manufacturer claims that only 4% of his products supplied by him are defective. A random sample of 600 products contained 36 defectives. Test the claim of the manufacturer.	6	L4	4	3
	(b)	A random sample of 200 villages was taken from district A and average population per village was 485 with standard deviation 50. Another random sample of 250 villages from the same district gave an average population of 510 per village with standard deviation of 40. Is the difference between the averages of two sample statistically significant?	6	L4	4	4

Blooms's Taxonomy Levels
L1 – 10%, L2 – 5%, L3 – 53.75%, L4 – 28.75%, L5 – 2.5%.

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